

Soil Hydrology in Lodgepole Pine Ecosystems in Southeastern Wyoming

RAY C. NYBERG AND TIMOTHY J. FAHEY*

ABSTRACT

The rates and pathways of water movement through the soil were examined in several lodgepole pine (*Pinus contorta* ssp. *latifolia*) forests in southeastern Wyoming to improve our understanding of the dynamics of solute flux in forest ecosystems. We employed internal drainage and tracer pulse methods to determine hydraulic conductivity, and we examined snowmelt infiltration and drainage water chemistry on contrasting soils supporting lodgepole pine forests. Saturated hydraulic conductivity (K_s) values were lognormally distributed for 27 plots at each of two sites examined. The median value was significantly higher at a site on a gravelly, loamy-sand (median $K_s = 1.20$ cm/h) than on a bouldery, sandy-loam ($K_s = 0.49$). At these sites the high range in both K_s values and in the transit time for a Cl^- pulse were suggestive of highly variable water flow pathways through the soil. Early in the spring snowmelt period much

of the infiltrating water circumvented the soil matrix, probably flowing around entrapped air pockets and through macropores. Although large systematic differences in chemistry were not observed between soil solutions collected with tension and tensionless lysimeters, saturated flow (collected as drips from the ceiling of soil tunnels) contained about three times lower cation concentrations than the lysimeter solutions. To obtain more accurate estimates of nutrient leaching from the tree rooting zone better quantification of the chemistry of percolating soil water is needed.

Additional Index Words: Hydraulic conductivity, Infiltration, Leaching, Macropores, Spatial variation.

R.C. Nyberg, Dep. of Botany, Univ. of Wyoming, Laramie, WY 82071; and T.J. Fahey, Dep. of Natural Resources, Fernow Hall, Cornell Univ., Ithaca, NY 14853. Research supported by the National Science Foundation (DEB 81-13546, BSR 83-16924). Contribution from the Dep. of Botany, Univ. of Wyoming, and Dep. of Natural Resources, Cornell Univ. Received 26 May 1987. *Corresponding author.

Published in Soil Sci. Soc. Am. J. 52:844-849 (1988).

A BETTER UNDERSTANDING of the rates and pathways of water movement through soils is essential for modelling soil chemical reactions (Luxmoore et al., 1978) and estimating subsurface leaching in forest ecosystems (Fahey and Yavitt, in press). Many forest soils exhibit complex hydrologic behavior, including preferential flow pathways (DeVries and Chow, 1978) or a dual-pore system (Shaffer et al., 1979). Also, some very high flow rates have been observed in wet

soils (DeVries and Chow, 1978; Mosley, 1982). Transport via root channels and the high stoniness in many forest soils favor rapid drainage of water when impeding horizons are absent, and the interaction between this rapidly percolating water and the soil matrix may be limited.

The objective of the present study was to determine the rates and pathways of water movement through contrasting soils in lodgepole pine ecosystems. These forests dominate the montane landscape at elevations of about 2700 to 3100 m in the central Rocky Mountains, and the montane and subalpine watersheds supply water for agricultural, municipal and industrial uses in the valleys below. Lodgepole pine invades burned over sites on nearly all combinations of soil and topography in this region and despite the relatively uniform vegetation and climate, marked variation in forest hydrology has been observed (Knight et al., 1985). The hydrology of lodgepole pine ecosystems is dominated by the winter snowpack which begins to accumulate in October and melts in a 3- to 5-wk period in May or June. Little root zone leaching occurs in summer as soil moisture is depleted by tree uptake (Knight et al., 1985).

We examined soil hydrology in three intensive research sites in southeastern Wyoming, each dominated by lodgepole pine (Table 1). Soils included a gravelly, loamy-sand derived from a coarse, granitic regolith (Lithic Cryochrept at Albany site); a bouldery, clay-loam on a quartzite-rich glacial moraine (Typic Cryoboralf at French Creek site); and a bouldery, sandy-loam on mixed glacial till (Typic Cryoboralf at Nash Fork site). We also report observations from three other lodgepole pine ecosystems used in our earlier studies (Rock Creek, Dry Park, and Fox Park). More detailed descriptions of these and the intensive sites may be found elsewhere (Knight et al., 1985; Fahey and Yavitt, in press).

We expected high spatial variation in hydraulic conductivity (K) and soil water movement both within and among sites in these lodgepole pine ecosystems. As commonly observed in field soils, a lognormal distribution of K values was anticipated (Jury et al., 1982). Also, we expected occasionally to encounter zones of preferential flow with extremely high saturated hydraulic conductivity (K_s). Maximum rates of internal redistribution of soil water were expected to exceed the snowmelt rate, except in high clay soils where overland flow has been observed (Knight et al., 1985).

MATERIALS AND METHODS

Hydraulic Conductivity

We measured in situ hydraulic conductivity using an internal drainage approach (Hillel, 1982). Libardi et al. (1980) described a method which does not require the tensiometer data that are so difficult to obtain in bouldery glacial soils. Assuming an exponential relationship between K and volumetric water content (θ)

$$K(\theta) = K_0 \exp [\beta(\theta - \theta_0)] \quad [1]$$

and assuming negligible pressure head (minimal ponding depth), then K_0 and β can be calculated from the time course of soil drainage from θ_0 (θ after prolonged ponding) by

Table 1. Soil physical properties for three lodgepole pine ecosystems, southeastern Wyoming.

Site	Soil depth cm	Total porosity† %	Stones (>10 mm) — % by mass —	Gravel (2–10 mm) —	Silt % <2 mm fraction	Clay %	θ_{fc} ‡ cm
Albany	0–30	50	0	55	18	12	6
	30–60	7	0	73	6	8	
Nash Fork	0–30	47	20	31	33	15	9
	30–60	36	18	46	24	15	
French Creek	0–30	54	19	19	40	24	12
	30–60	45	16	49	29	20	

† Assuming particle density = 2.65.

‡ Volumetric water content at field capacity (0–80 cm).

regression of $\theta_0 - \theta$ vs. $\ln t$ as represented in a reformulation of the Richard's equation for large time (t)

$$\theta_0 - \theta = (1/\beta) \ln t + (1/\beta) \ln (K_0 \beta / a z) \quad [2]$$

In this equation a is obtained from the relationship of average water content (θ^*) to depth z , and θ at depth z

$$\theta^* = a\theta + b \quad [3]$$

as measured for each redistribution plot.

We employed this internal drainage method on 32 plots at the Nash Fork site and 34 plots at the Albany site. Neutron probe access holes were power augered to a depth of 1.0 to 1.8 m using a 6-cm bit. Aluminum irrigation pipe (5-cm diam.) was inserted in each hole, and soil from the holes was used to back fill firmly around the pipes.

Because the manufacturer's calibration of the neutron moisture meter (Campbell Scientific Inc., Logan, UT) may not be suitable for actual field determinations (Carneiro and DeJong, 1985), we calibrated the neutron probe in the laboratory. Soils from two of the research sites (Nash Fork, Albany) were packed to about field bulk density in 55-gallon cylindrical steel barrels (0.7-m diam. by 1.1-m height) equipped with Al access tubes. Water was added in measured increments to dry soil and allowed to equilibrate within the barrels before neutron probe readings were taken. Calibration curves for the two soils were not significantly different, but they differed from the manufacturer's calibration, possibly because of organic matter in these profiles.

Internal drainage measurements were carried out in October and November, 1983 when trees were dormant (e.g., Smith et al., 1984) to avoid problems with concurrent root water uptake (Hillel, 1982). Small plots (1.2 m diam.) around each neutron probe access tube were ponded with stream water until θ was nearly constant throughout the soil profile; typically this required 50 to 100 cm of added water. After steady state was reached, the plots were covered with plastic sheets to prevent evaporation, and neutron probe readings were taken at 20-cm depth increments one to three times each day for the next several days.

We also estimated the rate of soil water movement from the velocity of a pulse of chloride through wet soil. Distilled water (about 30 cm) was added to 1.5-m by 1.5-m plots in the Nash Fork, Albany, and French Creek sites. A 4-cm pulse of 100 mg/L NaCl solution was followed by addition of 8 to 10 cm/d of distilled water for 4 to 5 d. We collected soil solutions 8 to 10 times in each plot using one porous-cup tension lysimeter and one zero-tension funnel lysimeter placed at about 60-cm depth in each plot. Chloride was analyzed colorimetrically (Zall et al., 1956) on a Scientific Instruments, Inc. (Scientific Instruments, Inc., West Palm Beach, FL) continuous-flow analyzer (CFA). We used the time interval between application of the pulse and appearance of the chloride peak at the lysimeter to calculate water flux in wet soils.

Snowmelt Infiltration and Drainage

We measured concurrent changes in water content of the snowpack and the soil during the early phases of snowmelt to document the effectiveness of the upper mineral soil in retaining snowmelt. Ten 2-m by 2-m plots were established at the Albany and Fox Park sites. Four neutron probe access holes were augered and lined with pipes to permit access from above the snowpack. At 3-d intervals during the 1983 snowmelt period, we measured soil water content by neutron probe and snowpack water content with calibrated Montrose snow tubes.

Having observed dry pockets of soil during the snowmelt period, we systematically sampled for these nonwetted soils in three sites during late snowmelt, 1983. The remaining snowpack was excavated and soils carefully examined after exposing wedges to a depth of about 20 cm with a shovel on 150 random 1-m by 1-m plots at each site. The area and approximate volume of nonwetted soil was estimated visually, and ten samples each of wet and dry soil were collected for laboratory analysis. The distinction between water-saturated wet soil and dry soil was very obvious in most instances. Water content of wet and dry soil samples was measured gravimetrically. A wettability index was obtained from the contact angle of water droplets on soil surfaces, measured microscopically (Hillel, 1982).

Drainage Water Chemistry

We used zero-tension lysimeters to collect soil solutions at three sites (Albany, Nash Fork, Fox Park) during the snowmelt interval in 1983, for comparison with tension lysimeter samples which are reported in detail by Fahey and Yavitt (in press). Sand-filled funnel lysimeters (Turner and Johnson, 1985) and zero-tension lysimeters (Jordan, 1968) collected saturated flow at 2- to 4-d intervals throughout the snowmelt drainage period. No differences were observed between these two lysimeter types so henceforth both are denoted tensionless.

Saturated flow also was collected as drips from the ceilings of soil tunnels. Lysimeters and drip spots were accessible during snowmelt from sheltered soil trenches.

Each sample was filtered through prerinsed glass fiber filters (0.5 μm effective pore size) before being analyzed for major dissolved constituents. In this paper we report values for Ca^{2+} , K^+ , Na^+ and SO_4^{2-} , only. The cations were measured by atomic absorption spectrophotometry and SO_4^{2-} by

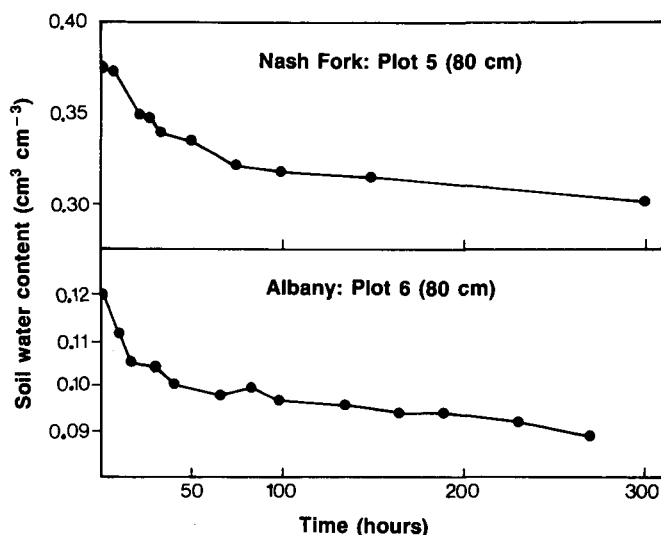


Fig. 1. Changes in soil water content (80-cm depth) following prolonged ponding for representative plots in two lodgepole pine forest sites.

a methylthymol blue method on the CFA (Anonymous, 1971).

RESULTS AND DISCUSSION

Hydraulic Conductivity and Soil Water Flux

In general, a consistent exponential decrease in soil water content was observed during drainage from the plots in both sites (Fig. 1); however, at the Nash Fork site a brief lag (2 to 6 h) in drainage was noted in several of the plots, probably indicating that perfect steady-state drainage was not achieved (R. J. Wagenet, 1987, personal communication). Values of the correlation coefficient for equation 2 (i.e., $\theta_o - \theta$ vs. $\ln t$) and Eq. [3] (θ^* vs. θ) were used to indicate the validity of the approximate solutions (Jones and Wagenet, 1984). Although all the correlations were highly significant ($p < 0.01$), four plots at Nash Fork and seven at Albany were eliminated from further consideration because less than half of the variance in $\theta_o - \theta$ was explained by the linear relationship with $\ln t$. Thus, the calculation of K_o by the θ method may not be appropriate (Jones and Wagenet, 1984).

Saturated hydraulic conductivity (K_o) values calculated from soil water retention data for 0- to 80-cm depth by the θ method of Libardi et al. (1980) were lognormal for two sites examined (Fig. 2), with most values clustered at the low end of the range. Mean and median values were about three times higher in a gravelly, loamy-sand at Albany than in a bouldery, sandy-loam at Nash Fork (Table 2), and these values were within the range of literature observations for these soil texture classes (Hillel, 1982). The large range in K_o values observed in these two sites is typical of field soils and suggests the existence of highly variable water flow pathways within these soils. This suggestion was further supported by observations of the movement of a chloride pulse (see below). In both sites the pulse moved from the surface to occasional drip spots on the ceiling of soil tunnels at 60-cm depth in 1 to 2 h,

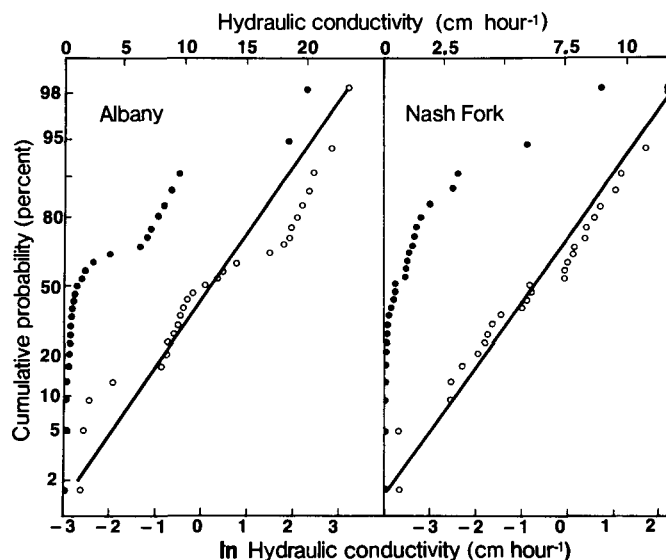


Fig. 2. Cumulative probability distribution of hydraulic conductivity (closed circles) and $\ln$ hydraulic conductivity (open circles) for two lodgepole pine sites estimated by the θ method of Libardi et al. (1980).

Table 2. Hydraulic characteristics of soils from lodgepole pine ecosystems.[†]

Site	Parameter	Chloride pulse	Internal drainage method	
		K_o	K_o	β
		cm/h		
Albany	Maximum	16	20.10	385.0
	Minimum	1.1	0.07	63.9
	Mean	6.0	4.16	179.7
	Median	2.8	1.20	166.0
Nash Fork	Maximum	3	9.00	185.2
	Minimum	0.3	0.03	8.4
	Mean	1.3	1.33	56.5
	Median	0.8	0.49	39.2
French Creek	Maximum	2		
	Minimum	0.03		
	Mean	0.6		
	Median	0.3		

[†] Saturated hydraulic conductivity (K_o) and exponential constant (β) for the relationship of K and θ , estimated by an internal drainage method (Libardi et al., 1980) and K_o calculated from the velocity of a chloride pulse through wet soil.

clearly evidence of macropore flow and illustrating the operation of a dual-pore drainage system in these forest soils (Shaffer et al., 1979). All but the very lowest K_o values observed in these soils were generally higher than the maximum rates of snowmelt in lodgepole pine ecosystems (Knight et al., 1985), and overland flow was never observed in either site during the snowmelt period.

The chloride-pulse method was used to estimate soil water flux in three lodgepole pine ecosystems, the Nash Fork and Albany sites discussed earlier and the French Creek site with a bouldery, clay-loam soil (Fig. 3). Median transit times were nearly five times longer for Nash Fork than for the coarse Albany soil, whereas for the fine soil at French Creek median transit time was over twice as long as for Nash Fork. Low flux rates and K_o values were anticipated for the fine French Creek soil where overland flow of snowmelt occurs each spring.

Assuming a unit hydraulic gradient and correcting for actual θ values, we calculated K_o from the transit time data. For both the Nash Fork and Albany sites, these K_o values were comparable to those calculated by the internal drainage method (Table 2). The median value for the French Creek site (0.3 cm/h) is comparable to the maximum snowmelt rates observed in this forest during the late stages of spring snowmelt (Knight et al., 1985) when small amounts of overland flow occur.

Retention of Snowmelt

Because dry summers and early snow accumulation usually result in dry soils (< -0.8 MPa) prior to snowmelt in these lodgepole pine ecosystems (Fahey and Knight, 1986), substantial air entrapment was anticipated during snowmelt infiltration the following spring. We measured the retention of snowmelt in the upper soil horizons of the lodgepole pine site at Fox Park in 1984 using the neutron probe method. Early in the snowmelt period significantly larger amounts of water had melted from the snowpack than appeared in the upper 80 cm of the soil profile. The maximum difference was about 11 cm on 29 Apr. 1984, when snow-

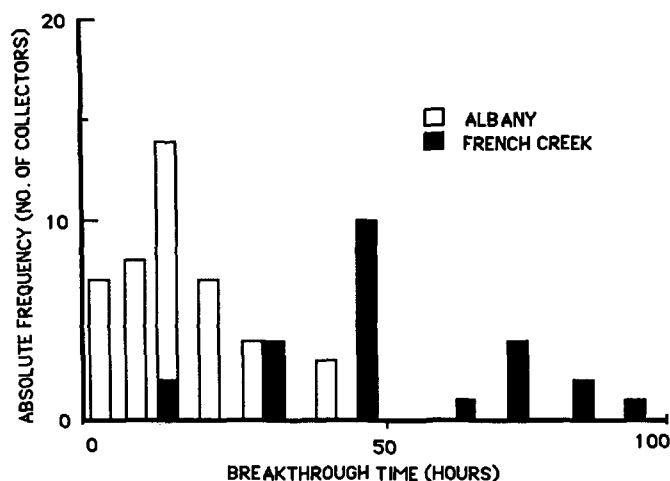


Fig. 3. Histograms of transit time for movement of a pulse of Cl^- to about the 60-cm depth in two lodgepole pine sites.

melt totalled 13.5 cm but the change in θ was only 2.3 cm. This result suggests that much of the melting snow was not retained in the soil matrix but rather was moving rapidly through macropores or circumventing entrapped air pockets. This profile retains about 8 cm at field capacity, and no significant overland flow or sublimation of snow occurs (Knight et al., 1985). However, our visual observations did suggest that some of the channelling could have resulted from nonuniform melting from the base of the snowpack where an ice-covered layer had developed.

To corroborate these observations we excavated the remaining snowpack at 150 or 200 random locations in three lodgepole pine sites (Albany, Fox Park, Nash Fork). The frequency and dimensions of dry soil pockets on 0.5-m by 0.5-m plots are summarized in Table 3. Even following the melt of over 30 cm of snow, about 15% of the soil at the Nash Fork and Fox Park sites had not wetted. These dry soil pockets were non-wettable based upon the criterion of water droplet contact angles (Hillel, 1982) which were much higher than for the soil samples that had been wetted during snowmelt. Thus, these soils apparently contained chemical compounds conferring hydrophobic behavior which, in conjunction with macropore flow and air entrapment, help explain the restricted water retention observed during snowmelt.

Other evidence of snowmelt retention characteristics was obtained inadvertently with porous-cup lysimeters used to sample soil solutions during the spring flush period (Fahey and Yavitt, in press). At the Fox Park, Nash Fork and Dry Park sites, 20% to 30% of the total of 60 lysimeters (at 40-cm depth) failed to collect solutions each spring until the local perched

Table 3. Frequency and size distribution of dry soil pockets near the end of the snowmelt period in three lodgepole pine forest sites in southeastern Wyoming.

Site	No. samples	Total frequency	Frequency by size class, cm ²			
			<10	10-30	30-500	>500
			%			
Albany	150	1.3	0	0	1.3	0
Nash Fork	150	14.0	3.3	2.7	4.0	4.0
Fox Park	200	16.5	1.5	2.0	7.0	6.0

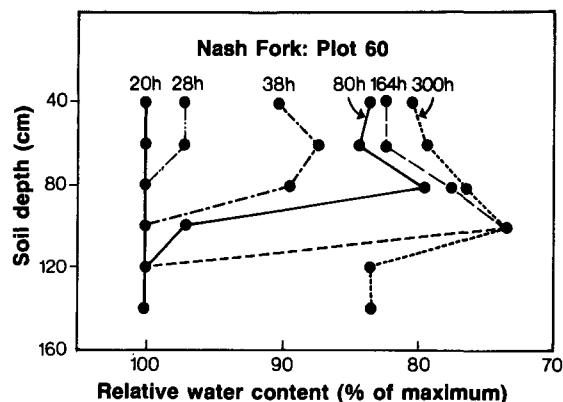


Fig. 4. Drainage of a saturated profile following snowmelt on a representative plot in the Nash Fork site in spring, 1984.

water table rose to the corresponding depth in the soil (based upon soil trenches). Apparently snowmelt water was channelled around these lysimeters probably via macropores. Similar results were obtained with tensionless lysimeters.

Snowmelt Drainage

The gentle topography which characterizes many areas of lodgepole pine ecosystems in southeastern Wyoming leads to restricted rates of transmission of snowmelt and to the rise of a seasonal, local perched water table in many areas. The rate of fall of the water table, as gauged with soil trenches, ranged from 3 to 13 cm/d among three sites exhibiting this phenomenon.

At the Nash Fork site no water table was observed, and following snowmelt in 1984, internal drainage was measured in the same manner as for the ponding experiments. A large range and lognormal distribution were observed for these values as well, but the magnitude of the values could be affected by tree root uptake at this time (Fahey, 1979a). In 7 of the 33 plots measured, slow drainage of a supersaturated soil profile was observed (Fig. 4), and the rate of fall of this drainage front averaged about 0.4 cm/h.

At the French Creek site (fine soil), drainage of snowmelt water was gradual (0.12 cm/h) and spatially uniform. All six neutron probe holes gave similar results. At this site and at another site with fine soil (loamy-clay at Rock Creek; Fahey and Yavitt, in press) overland flow was observed during snowmelt, and both of these sites exhibited saturated conditions despite the moderate slope (10 to 20%) of the terrain. In the landscape around the Rock Creek site overland flow is a major pathway of hydrologic transport during snowmelt (Fahey, 1979b), whereas it is a minor component at French Creek (Knight et al., 1985). Thus, for gentle and moderate slopes in this ecosystem, overland flow apparently becomes important when soil clay content exceeds that at French Creek (20–24% clay).

Significant differences in chemistry usually were not observed between soil solutions collected by tension (porous cup) and tensionless lysimeters in two sites. For example Ca^{2+} concentrations at Albany were slightly ($p > 0.05$) higher for tensionless (0.21 mg/L) than tension lysimeter samples (0.16 mg/L), whereas

the reverse was true at Nash Fork (0.26 vs. 0.29 mg/L). Differences for Na^+ and SO_4^{2-} were even smaller, whereas K^+ concentrations were significantly ($p < 0.05$) higher for tensionless lysimeters at Albany but nearly identical for both types at Nash Fork.

In contrast, concentrations of all these solutes were much lower ($p < 0.01$) in saturated flow collected as drips from the ceilings of soil tunnels. For example, Ca^{2+} and K^+ averaged three times higher in lysimeter samples than in drip samples. We speculate that some rapidly flowing water, collected as drips from soil tunnels, interacts little with the soil matrix and that solutions collected with lysimeters represent largely the more slow-moving, interstitial water. That drips were observed intermittently only in 8 of the 24 tunnels in the Nash Fork and Albany stands suggests that such rapid flow may not be the dominant pathway of soil water movement. As indicated by their similarity to the tension lysimeter solutions, apparently the tensionless lysimeters did not collect significant amounts of this rapidly-flowing water.

CONCLUSIONS

Soil hydrology in these lodgepole pine ecosystems is characterized by high spatial variation both within and between sites. Within-site variation, which approximated a lognormal distribution, provides evidence for an array of flow pathways with different properties, probably owing to the complex three-dimensional structure of these stony soils. Some snowmelt appears to be channelled along preferential flow pathways and much of the soil may remain unwetted during early stages of the snowmelt period. Soil clay content appears critical in defining between-site differences in soil hydrology; below about 20% clay, hydraulic conductivity usually exceeds the snowmelt rate, whereas at higher clay contents (over 28% clay) drainage is constrained and overland flow becomes prominent. The rise of a local perched water table during late snowmelt plays an important role on sites with gentle topography and/or fine soils. These hydrologic phenomena certainly affect soil chemical reactions and solute transport from the soils. Relatively immobile soil water may interact only sparingly with the mobile phase, and because tension lysimeters may collect primarily from the immobile phase (Shaffer et al., 1979), estimates of solute flux based on these solution samples and a simple cascade-type hydrology model (Running, 1984) may be inaccurate. More accurate estimates of root-zone leaching will require sampling of soil solutions with a combination tension and tensionless lysimeters of different designs and additional evaluation of the proportions of percolating soil water with solute concentrations represented by the solutions obtained with the different types of collector.

REFERENCES

- Anonymous. 1971. Technicon industrial systems. Sulfate in water and waste water, method A411, no. 118-71, Technicon Inst., Tarrytown, NY.
- Carneiro, C., and E. DeJong. 1985. In situ determination of the slope of the calibration curve of a neutron probe using a volumetric technique. *Soil Science* 139:250–254.
- DeVries, J., and T.L. Chow. 1978. Hydrologic behavior of a forested mountain soil in coastal British Columbia. *Water Resources Res.*

- 14:935-942.
- Fahey, T.J. 1979a. The effect of night frost on the transpiration of *Pinus contorta* ssp. *latifolia*. *Oecol. Planta.* 14:483-490.
- Fahey, T.J. 1979b. Changes in the nutrient content of snow water during outflow from Rocky Mountain coniferous forest. *Oikos* 32:422-428.
- Fahey, T.J., and D.H. Knight. 1986. Lodgepole pine ecosystems. *BioScience* 36:610-617.
- Fahey, T.J., and J.B. Yavitt. Soil solution chemistry in lodgepole pine (*Pinus contorta* ssp. *latifolia*) ecosystems, southeastern Wyoming, USA. *Biogeochemistry* (in press).
- Hillel, D. 1982. Introduction to soil physics. Academic Press, Inc., Orlando, FL.
- Jones, A.J., and R.J. Wagenet. 1984. *In situ* estimation of hydraulic conductivity using simplified measures. *Water Resources Res.* 30:1620-1626.
- Jordan, C.F. 1968. A simple, tension-free lysimeter. *Soil Sci.* 105:81-86.
- Jury, W.A., L.H. Stolzy, and P. Shouse. 1982. A field test of the transfer function model for predicting solute transport. *Water Resour. Res.* 18:369-376.
- Knight, D.H., T.J. Fahey, and S.W. Running. 1985. Water and nutrient outflow from contrasting lodgepole pine forests in Wyoming. *Ecol. Monogr.* 55:29-48.
- Libardi, P.L., K. Reichardt, D.R. Nielsen, and J.W. Biggar. 1980. Simple field methods for estimating soil hydraulic conductivity. *Soil Sci. Soc. Am. J.* 44:3-7.
- Luxmoore, R.J., C.L. Begovich, and K.R. Dixon. 1978. Modelling solute uptake and incorporation into vegetation and litter. *Ecol. Model.* 5:137-171.
- Mosley, M.P. 1982. Subsurface flow velocities through selected forest soils. South Island, New Zealand. *J. Hydrol.* 55:65-92.
- Running, S.W. 1984. Documentation and preliminary validation of H2OTRANS and DAYTRANS, two models for predicting transpiration and water stress in western coniferous forests. USDA-FS Res. Pap. RM-252. Rocky Mountain For. Range Exp. Stn., Fort Collins, CO 80526.
- Shaffer, K.A., D.D. Fritton, and D.E. Baker. 1979. Drainage water sampling in a wet, dual-pore soil system. *J. Environ. Qual.* 8:241-246.
- Smith, W.K., D.R. Young, G.A. Carter, J.L. Hadley, and G.M. McNaughton. 1984. Autumn stomatal closure in six conifer species of the Central Rocky Mountains. *Oecologia* 63:237-242.
- Turner, R.A., A.H. Johnson, and D. Wang. 1985. Biogeochemistry of aluminum in the McDonald's Branch watershed, New Jersey. *J. Environ. Qual.* 14:314-323.
- Zall, D.M., D. Fisher, and M.D. Garner. 1956. Photometric determination of chloride in water. *Anal. Chem.* 28:1665-1668.